

Python Data Visualization Assignment Report

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1 Introduction & Dataset

The aim of this project is to explore data visualization using Python and different Python libraries. The assignment prompt allowed for several Python libraries to be used including:

- Matplotlib
- ipywidgets
- Plotly
- Bokeh
- Jupyter Dash
- Seaborn

1.1 Dataset

Data for this project was required to not have been selected by a different student. To minimize the risk of this happening, activity data from a Garmin Forerunner 245 was chosen, reducing the chances of another student having the same dataset.

The data set, located in `./data/activities.csv`, consists of:

- 31 columns & 180 rows.
- Data spanning from: 21 February 2021 - 30 March 2026 (5 years).
- Varying data types (integers, floats, dates, time spans, locations, strings).

2 Data Processing & EDA

3 Visualization Process

4 Research Questions

4.1 Question 1

4.2 Question 2

4.3 Question 3

5 Visualization Summary

Question	Charts	Type	Library	Screenshot
X	Image X	Bar	Seaborn	Fig. TODO

Table 1: Requirements fulfillment summary

6 Visualizations

6.1 Chart 1

Figure 1: Your caption

7 Conclusions

AI Usage Declaration

Bibliography